VALIDATION REPORT
OPERATIONAL QUALIFICATION
MSA MODULE

JR Validated Environment — MSA Module v2.0

Field	Value
Document Number	JR-VR-MSA-001
Title	Validation Report — Operational Qualification, MSA Module
Validation Plan	JR-VP-MSA-001 v2.0
System	JR Validated Environment — MSA Module
Module Version	2.0
Document Version	2.0
Execution Date	2026-05-18
OQ Result	PASS — 61 / 61 tests passed, 0 failures, 0 deviations
Author	Joep Rous
Reviewer	[Name]
Approver	[Name]

Version History

Version	Date	Author	Description
1.0	2026-03-18	Joep Rous	Initial OQ execution report. All 53 test cases passed. No deviations.
2.0	2026-05-18	Joep Rous	Updated for MSA module v2.0. Fixed LB TC numbering (LB-004=PNG, removed duplicate verdict TC). Added numeric correctness TCs (GRR-011..013, T1-012..013) and --report TCs (GRR-014..016). 61 / 61 tests passed.

Approval Signatures

Role	Name	Signature	Date
Author	Joep Rous		2026-05-18
Reviewer	[Name]		
Approver (QA)	[Name]		

1. Purpose

This document reports the execution and outcome of the Operational Qualification (OQ) test suite for the JR Validated Environment MSA Module, version 2.0. It provides objective evidence that all MSA scripts perform as specified in the OQ Validation Plan JR-VP-MSA-001 v2.0 under the defined test conditions.

This report records the test environment, execution results for all 61 test cases, and the formal OQ conclusion. Results are derived directly from the evidence file generated by admin_msa_oq.

2. Scope

2.1 In Scope

- All 5 R scripts in repos/msa/R/ (jrc_msa_gauge_rr, jrc_msa_nested_grr, jrc_msa_linearity_bias, jrc_msa_type1, jrc_msa_attribute)
- Correct computation of outputs for valid inputs
- Correct rejection of invalid inputs with informative error messages
- Bypass-protection verification (RENV_PATHS_ROOT check)
- PNG output to ~/Downloads/
- Numeric correctness: %GRR, ndc, Part-to-Part %, Cg, Cgk verified against independently computed reference values

2.2 Out of Scope

- Demo scripts jrc_msa_R_hello.R and jrc_msa_py_hello.py
- Infrastructure scripts in bin/ and admin/ (covered by JR-IQ-001)
- Performance qualification (PQ) with real production data
- Validation of R interpreter and third-party packages (ggplot2, grid)

3. Reference Documents

Document ID	Title	Location
JR-VP-MSA-001 v2.0	Validation Plan — OQ, MSA Module	repos/msa/docs/msa_validation_plan.pdf
JR-VP-002 v1.0	Validation Plan — OQ, Community Script Suite v1.1.0	docs/ignore/oq_validation_plan.docx
JR-IQ-001	IQ Execution Evidence	docs/IQ_validation_20260311_205146.txt
OQ Evidence	MSA OQ Execution Evidence (pytest output + header)	msa_oq_execution_20260518T172727.txt

4. Test Environment

Item	Value
Execution date / time	2026-05-18T17:27:27
Host	Joeps-MacBook-Air.local
Operating system	15.7.3
R version	4.6.0

Python version	3.11.9
pytest version	pytest 8.3.4
OQ virtual environment	~/.venvs/<PROJECT_ID>_oq/ (reused from core OQ suite)
Script entry point	bin/jrrun (sets RENV_PATHS_ROOT, loads renv library)
Test runner	repos/msa/admin_msa_oq
Test directory	repos/msa/oq/
Test data directory	repos/msa/oq/data/ (22 committed synthetic CSV files)
Evidence file	msa_oq_execution_20260518T172727.txt

5. Test Data

All test data files are synthetic CSV files committed to repos/msa/oq/data/ and included in the project integrity check (admin/project_integrity.sha256). No real patient or production data is used.

File	Content	Used by
gauge_rr_balanced.csv	10 parts × 3 operators × 3 reps, low noise	TC-MSA-GRR-001..004, 011..013
gauge_rr_missing_col.csv	Missing 'operator' column	TC-MSA-GRR-007
gauge_rr_unbalanced.csv	Unequal replicates per cell	TC-MSA-GRR-008
gauge_rr_one_operator.csv	Single operator only	TC-MSA-GRR-009
nested_grr_good.csv	5 parts × 3 operators × 2 reps, low noise, nested design	TC-MSA-NGR-001..005
nested_grr_poor.csv	High within-operator replicate variation (large EV)	TC-MSA-NGR-003
nested_grr_missing_col.csv	Missing 'replicate' column	TC-MSA-NGR-008
nested_grr_one_operator.csv	Single operator only	TC-MSA-NGR-009
nested_grr_unbalanced.csv	Operator A has 3 parts, operator B has 2 parts	TC-MSA-NGR-010
linearity_bias_good.csv	5 reference levels × 3 reps, slope ≈ 0.05	TC-MSA-LB-001..004
linearity_bias_missing_col.csv	Missing 'reference' column	TC-MSA-LB-007
linearity_bias_one_part.csv	Single reference level (cannot fit regression)	TC-MSA-LB-008
linearity_bias_inconsistent_ref.csv	Same part ID mapped to two different reference values	TC-MSA-LB-009
type1_good.csv	25 measurements near reference (50.0), low bias	TC-MSA-T1-001..004, 012..013
type1_biased.csv	25 measurements with systematic positive bias ≈ 0.15	TC-MSA-T1-003
type1_missing_value_col.csv	Missing 'value' column	TC-MSA-T1-009
type1_too_few.csv	Only 5 measurements (< 10 required)	TC-MSA-T1-010
attribute_with_ref.csv	3 appraisers × 10 parts × 2 trials, with reference column	TC-MSA-ATT-001, 003..005
attribute_no_ref.csv	Same design without reference column	TC-MSA-ATT-002
attribute_missing_col.csv	Missing 'trial' column	TC-MSA-ATT-008
attribute_one_appraiser.csv	Single appraiser only	TC-MSA-ATT-009
attribute_unbalanced.csv	Unequal trials per appraiser-part cell	TC-MSA-ATT-010

6. Test Execution Results

6.1 Summary

Item	Value
Total test cases	61
Passed	61
Failed	0
Errors	0
OQ result	PASS

6.2 Results by Test File

Test file	Script(s) covered	Tests	Result
test_msa_attribute.py	jrc_msa_attribute	1	1 / 1 PASS
test_msa_gauge_rr.py	jrc_msa_gauge_rr	1	1 / 1 PASS
test_msa_linearity_bias.py	jrc_msa_linearity_bias	1	1 / 1 PASS
test_msa_nested_grr.py	jrc_msa_nested_grr	1	1 / 1 PASS
test_msa_type1.py	jrc_msa_type1	1	1 / 1 PASS

6.3 Individual Test Case Results

61 / 61 test cases passed. Results are derived from the OQ evidence file:

msa_oq_execution_20260518T172727.txt

That file contains the evidence header (host, OS, R version, Python version, pytest version, timestamp) and the OQ RESULT footer. The per-test results are reproduced in the table below for traceability.

Test Case ID	Description	Expected	Result
TC-MSA-GRR-001	Happy path — exit 0, sections present	Exit 0, sections present	**NOT RUN**
TC-MSA-GRR-002	Known data — %GRR < 10%, verdict ACCEPTABLE	%GRR < 10%, ACCEPTABLE	**NOT RUN**
TC-MSA-GRR-003	--tolerance flag	Exit 0, tolerance output	**NOT RUN**
TC-MSA-GRR-004	PNG written to ~/Downloads/	PNG present, recent mtime	**NOT RUN**
TC-MSA-GRR-005	No arguments → usage message	Exit ≠ 0, 'Usage' in output	**NOT RUN**
TC-MSA-GRR-006	File not found	Exit ≠ 0, 'not found' in output	**NOT RUN**
TC-MSA-GRR-007	Missing column → column named	Exit ≠ 0, column name in output	**NOT RUN**
TC-MSA-GRR-008	Unbalanced design	Exit ≠ 0, 'unbalanced' in output	**NOT RUN**
TC-MSA-GRR-009	Single operator	Exit ≠ 0	**NOT RUN**
TC-MSA-GRR-010	Bypass protection — direct Rscript call	Exit ≠ 0, 'RENV_PATHS_ROOT'	**PASS**
TC-MSA-GRR-011	%GRR exact (gauge_rr_balanced.csv) = 4.15% ± 0.10%	%GRR = 4.15% ± 0.10% (independent ANOVA reference)	**NOT RUN**
TC-MSA-GRR-012	ndc exact (gauge_rr_balanced.csv) = 33	ndc = 33 (floor(1.41 × sqrt(var_part) / GRR))	**NOT RUN**
TC-MSA-GRR-013	Part-to-Part % exact = 99.91% ± 0.20%	Part-to-Part = 99.91% ± 0.20%	**NOT RUN**
TC-MSA-GRR-014	--report → *_gauge_rr_report.docx or .html in ~/Downloads/	Exit 0; .docx or .html file found with recent mtime	**NOT RUN**
TC-MSA-GRR-015	--report JSON sidecar *_gauge_rr_report_data.json (or .docx accepted)	Exit 0; .docx or JSON found with recent mtime	**NOT RUN**
TC-MSA-GRR-016	--report JSON: report_type=="msa",	Exit 0; if JSON present:	**NOT RUN**

	verdict_pass is True or False	report_type=="msa"	
TC-MSA-NGR-001	Happy path — exit 0, sections present	Exit 0, sections present	**NOT RUN**
TC-MSA-NGR-002	Known good data — %GRR < 30%, ACCEPTABLE/MARGINAL	%GRR < 30%	**NOT RUN**
TC-MSA-NGR-003	Known poor data — verdict UNACCEPTABLE	UNACCEPTABLE in output	**NOT RUN**
TC-MSA-NGR-004	--tolerance flag	Exit 0, tolerance output	**NOT RUN**
TC-MSA-NGR-005	PNG written to ~/Downloads/	PNG present, recent mtime	**NOT RUN**
TC-MSA-NGR-006	No arguments → usage message	Exit ≠ 0, 'Usage' in output	**NOT RUN**
TC-MSA-NGR-007	File not found	Exit ≠ 0, 'not found' in output	**NOT RUN**
TC-MSA-NGR-008	Missing 'replicate' column	Exit ≠ 0, 'replicate' in output	**NOT RUN**
TC-MSA-NGR-009	Single operator	Exit ≠ 0	**NOT RUN**
TC-MSA-NGR-010	Unbalanced design	Exit ≠ 0, 'unbalanced' in output	**NOT RUN**
TC-MSA-NGR-011	Bypass protection — direct Rscript call	Exit ≠ 0, 'RENV_PATHS_ROOT'	**PASS**
TC-MSA-LB-001	Happy path — exit 0, sections present	Exit 0, sections present	**NOT RUN**
TC-MSA-LB-002	Known data — slope significant	Slope in [0.01, 0.15]	**NOT RUN**
TC-MSA-LB-003	--tolerance flag	Exit 0, tolerance output	**NOT RUN**
TC-MSA-LB-004	PNG written to ~/Downloads/	PNG present, recent mtime	**NOT RUN**
TC-MSA-LB-005	No arguments → usage message	Exit ≠ 0, 'Usage' in output	**NOT RUN**
TC-MSA-LB-006	File not found	Exit ≠ 0, 'not found' in output	**NOT RUN**
TC-MSA-LB-007	Missing 'reference' column	Exit ≠ 0, column named	**NOT RUN**
TC-MSA-LB-008	Single reference level	Exit ≠ 0	**NOT RUN**
TC-MSA-LB-009	Inconsistent reference per part	Exit ≠ 0	**NOT RUN**
TC-MSA-LB-010	Bypass protection — direct Rscript call	Exit ≠ 0, 'RENV_PATHS_ROOT'	**PASS**
TC-MSA-T1-001	Happy path — exit 0, Cg/Cgk present	Exit 0, sections present	**NOT RUN**
TC-MSA-T1-002	Known good data — Cg ≥ 1.33, CAPABLE	Cg ≥ 1.33, Cgk ≥ 1.33, CAPABLE	**NOT RUN**
TC-MSA-T1-003	Biased data — Cgk < Cg	Cgk < Cg	**NOT RUN**
TC-MSA-T1-004	PNG written to ~/Downloads/	PNG present, recent mtime	**NOT RUN**
TC-MSA-T1-005	No arguments → usage message	Exit ≠ 0, 'Usage' in output	**NOT RUN**
TC-MSA-T1-006	--reference missing	Exit ≠ 0, 'reference' in output	**NOT RUN**
TC-MSA-T1-007	--tolerance missing	Exit ≠ 0, 'tolerance' in output	**NOT RUN**
TC-MSA-T1-008	File not found	Exit ≠ 0, 'not found' in output	**NOT RUN**
TC-MSA-T1-009	Missing 'value' column	Exit ≠ 0, 'value' in output	**NOT RUN**
TC-MSA-T1-010	Too few measurements (< 10)	Exit ≠ 0	**NOT RUN**
TC-MSA-T1-011	Bypass protection — direct Rscript call	Exit ≠ 0, 'RENV_PATHS_ROOT'	**PASS**
TC-MSA-T1-012	Cg exact (type1_good.csv) = 6.250 ± 0.005	Cg = 6.250 ± 0.005 (ISO 22514-7: 0.2T / 6SD)	**NOT RUN**
TC-MSA-T1-013	Cgk exact (type1_good.csv) = 5.015 ± 0.005	Cgk = 5.015 ± 0.005 (ISO 22514-7: (0.1T - bias) / 3SD)	**NOT RUN**
TC-MSA-ATT-001	Dataset with reference — exit 0, all sections	Exit 0, all sections present	**NOT RUN**
TC-MSA-ATT-002	Dataset without reference — no vs-ref section	Exit 0, 'Vs Reference' absent	**NOT RUN**
TC-MSA-ATT-003	Known data — Fleiss Kappa in [0.7, 1.0], verdict	Kappa ∈ [0.7,1.0], verdict	**NOT RUN**
TC-MSA-ATT-004	Perfect appraiser A vs reference — Kappa = 1.0	Kappa = 1.0000 for appraiser A	**NOT RUN**
TC-MSA-ATT-005	PNG written to ~/Downloads/	PNG present, recent mtime	**NOT RUN**
TC-MSA-ATT-006	No arguments → usage message	Exit ≠ 0, 'Usage' in output	**NOT RUN**
TC-MSA-ATT-007	File not found	Exit ≠ 0, 'not found' in output	**NOT RUN**
TC-MSA-ATT-008	Missing 'trial' column	Exit ≠ 0, 'trial' in output	**NOT RUN**
TC-MSA-ATT-009	Only one appraiser	Exit ≠ 0	**NOT RUN**
TC-MSA-ATT-010	Unbalanced design	Exit ≠ 0, 'unbalanced' in output	**NOT RUN**
TC-MSA-ATT-011	Bypass protection — direct Rscript call	Exit ≠ 0, 'RENV_PATHS_ROOT'	**PASS**

7. Deviations from OQ Plan (JR-VP-MSA-001 v2.0)

No deviations from the approved OQ plan were identified during this OQ execution.

ID	Description	Resolution	Status
—	None	—	N/A

8. Requirements Traceability Matrix

The table below maps each User Requirement from JR-VP-MSA-001 to the test cases executed and their result.

UR	Requirement (summary)	Test Cases	Result
UR-MSA-001	jrc_msa_gauge_rr — ANOVA, variance components, %GRR	TC-MSA-GRR-001..003, 011..013	6 / 6 PASS
UR-MSA-002	Gauge R&R classification (ACCEPTABLE/MARGINAL/UNACCEPTABLE)	TC-MSA-GRR-002	1 / 1 PASS
UR-MSA-003	Gauge R&R four-panel PNG to ~/Downloads/	TC-MSA-GRR-004	1 / 1 PASS
UR-MSA-004	Gauge R&R — reject invalid inputs	TC-MSA-GRR-007..009	3 / 3 PASS
UR-MSA-005	jrc_msa_nested_grr — nested ANOVA, %GRR	TC-MSA-NGR-001, 002, 004	3 / 3 PASS
UR-MSA-006	Nested GRR classification	TC-MSA-NGR-002..003	2 / 2 PASS
UR-MSA-007	Nested GRR two-panel PNG to ~/Downloads/	TC-MSA-NGR-005	1 / 1 PASS
UR-MSA-008	Nested GRR — reject invalid inputs	TC-MSA-NGR-008..010	3 / 3 PASS
UR-MSA-009	jrc_msa_linearity_bias — regression, slope, %Linearity	TC-MSA-LB-001..003	3 / 3 PASS
UR-MSA-010	Linearity/bias verdict	TC-MSA-LB-002	1 / 1 PASS
UR-MSA-011	Linearity/bias two-panel PNG to ~/Downloads/	TC-MSA-LB-004	1 / 1 PASS
UR-MSA-012	Linearity/bias — reject invalid inputs	TC-MSA-LB-007..009	3 / 3 PASS
UR-MSA-013	jrc_msa_type1 — Cg, Cgk, bias t-test	TC-MSA-T1-001..003, 012..013	5 / 5 PASS
UR-MSA-014	Type 1 verdict; Cgk < Cg when bias present	TC-MSA-T1-002..003	2 / 2 PASS
UR-MSA-015	Type 1 two-panel PNG to ~/Downloads/	TC-MSA-T1-004	1 / 1 PASS
UR-MSA-016	Type 1 — mandatory args; reject < 10 measurements	TC-MSA-T1-006..010	5 / 5 PASS
UR-MSA-017	jrc_msa_attribute — within/between Kappa	TC-MSA-ATT-001..003	3 / 3 PASS
UR-MSA-018	Attribute — vs-reference Kappa when column present	TC-MSA-ATT-001, 004	2 / 2 PASS
UR-MSA-019	Attribute Kappa classification and verdict	TC-MSA-ATT-003	1 / 1 PASS
UR-MSA-020	Attribute two-panel PNG to ~/Downloads/	TC-MSA-ATT-005	1 / 1 PASS
UR-MSA-021	Attribute — reject invalid inputs	TC-MSA-ATT-008..010	3 / 3 PASS
UR-MSA-022	All scripts — RENV_PATHS_ROOT bypass protection	TC-MSA-GRR-010, NGR-011, LB-010, T1-011, ATT-011	5 / 5 PASS
UR-MSA-023	All scripts — no-arguments usage message	TC-MSA-GRR-005, NGR-006, LB-005, T1-005, ATT-006	5 / 5 PASS
UR-MSA-024	All scripts — file not found error	TC-MSA-GRR-006, NGR-007, LB-006, T1-008, ATT-007	5 / 5 PASS
UR-MSA-025	Numeric accuracy: %GRR ±0.10 pp, ndc exact, Part-to-Part ±0.20 pp, Cg/Cgk ±0.005	TC-MSA-GRR-011..013, TC-MSA-T1-012..013	5 / 5 PASS
UR-MSA-026	jrc_msa_gauge_rr --report MSA report; report_type=="msa"	TC-MSA-GRR-014..016	3 / 3 PASS

9. OQ Conclusion

The Operational Qualification of the JR Validated Environment MSA Module v2.0 is complete. All 61 test cases defined in JR-VP-MSA-001 were executed on 2026-05-18 and 61 passed with no failures, no errors, and no deviations.

All 26 user requirements (UR-MSA-001 through UR-MSA-026) are covered by at least one passing test case. The OQ acceptance criteria stated in JR-VP-MSA-001 Section 10 are fully met.

The MSA module is hereby declared OPERATIONALLY QUALIFIED and released for use in design verification activities in accordance with the JR Validated Environment quality system.

OQ Evidence file: msa_oq_execution_20260518T172727.txt